Narrative Structure Preservation in Cross-Cultural Story Retellings by Large Language Models

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Abstract

Storytelling is a fundamental form of human expression, often characterized by recurring structures such as conflict, transformation, and resolution. This study investigates whether such narrative structures can be computationally modeled and compared across cultures, and whether large language models (LLMs) preserve these structures when generating retellings in the same culture and over other cultures.

A corpus of folktales from three cultural traditions—Persian, German, and Native American—was compiled, alongside LLM-generated retellings of each story. Narratives were processed to extract key events as subject–verb–object triples and represented as directed graphs capturing their temporal progression. Narrative similarity was then evaluated using semantic embeddings and cosine similarity.

The proposed framework enables structural comparison of narratives beyond surface-level text similarity. The findings provide insights into cross-cultural narrative patterns and the extent to which LLMs preserve the structural logic of traditional stories.

Introduction

Storytelling is one of the most universal forms of human communication, used across cultures to transmit knowledge, values, and collective memory. Despite differences in language and tradition, many narratives share common structural elements such as conflict, transformation, and resolution. These recurring patterns suggest that stories can be understood not only through their content but also through their underlying sequence of events [1, 2].

With the rapid development of large language models (LLMs), machines are now capable of generating coherent narratives and retelling traditional stories [3, 4]. Although these models can effectively reproduce the language of storytelling, it remains unclear whether they also capture the deeper structural organization that characterizes human narratives.

Recent advances in natural language processing have enabled the transformation of textual narratives into structured representations that support computational analysis. In particular, modeling stories as sequences of events provides a way to compare narratives based on their progression of actions rather than their surface wording [5].

In this work, narratives are represented as event-based structures and analyzed to explore similarities across cultural traditions and between human-authored and machine-generated stories. The study aims to investigate whether common narrative patterns can be identified across diverse datasets and whether such structures are preserved in LLM-generated retellings.

Research Question

The observations explained in the introduction motivate the central research question of this study: can narrative structures be identified and compared across cultures by modeling stories as sequences of events, and do large language models preserve these structures when generating retellings?

Methodology

Corpus Design and Data Collection

The dataset consists of folktales from Persian, German, and Native American traditions, selected to ensure cultural diversity. Since many stories were translated into English, some linguistic nuances may be lost. Additionally, variation in narrative length and storytelling purpose may influence structural representation and comparison.

In addition to the human-written corpus, a complementary dataset of LLM-generated narratives was constructed. For each original story, a corresponding retelling was generated in the same culture using a large language model, with prompts designed to preserve the main characters and events while allowing variation in linguistic expression.

Moreover, cross-cultural narrative generation was performed to evaluate how narrative structures are adapted across cultural contexts. A fixed mapping was defined to systematically assign each source culture to a different target culture:

German → Persian, Persian → Native American, and Native American → German.

This mapping ensures that each narrative is retold within a distinct cultural framework, enabling controlled comparison between same-culture and cross-cultural generation conditions.

Narrative Structure Modeling

To enable structural comparison between narratives, each story is transformed into a graph-based representation that captures its sequence of events. Instead of relying on lexical similarity, this approach models narratives as ordered sequences of actions, allowing the analysis to focus on structural patterns such as progression and transformation.

The process begins with event extraction from the raw text using the spaCy natural language processing framework [6]. Each sentence is parsed to identify its root verb, which represents the main action. Dependency relations are then used to extract the associated subject and object, resulting in a simplified representation of each event as a subject–verb–object (SVO) triple [7]. Events are represented as subject–verb–object triples, capturing core action semantics. For example, the sentence "the hero defeated the monster" is normalized as:

hero | defeat | monster

These event triples are automatically extracted using dependency labels such as `nsubj` and `dobj`, with fallback rules for prepositional objects. Each event is stored along with metadata including its sentence position and original text.

The extracted events are then organized into a directed graph, where each node represents an event and edges connect consecutive events to reflect temporal order. Specifically, if event e_i precedes event e_{i+1}, a directed edge labeled `next` is added between them. This results in a graph structure that encodes the chronological progression of the narrative.

This representation is preferred over more complex narrative ontology approaches that explicitly model characters, events, and relations as separate components. While ontology-based methods can capture richer semantics, they typically require manual annotation and predefined schemas, making them less scalable for diverse, cross-cultural datasets [8].

In contrast, the event-graph approach provides a lightweight and fully automated alternative. By extracting events directly from syntactic structure and linking them temporally, the method captures the core action flow of the narrative without requiring manual labeling. This makes it well-suited for exploratory analysis across heterogeneous corpora.

Overall, representing narratives as event graphs preserves the temporal structure of storytelling while maintaining computational efficiency. This enables meaningful comparison between stories from different cultures, as well as between human-authored and LLM-generated narratives, by focusing on structural similarities rather than surface-level linguistic variation.

Evaluation Metrics

To evaluate the similarity between original narratives and their LLM-generated retellings, this study employs a combination of graph-based structural metrics and event-level evaluation measures. This dual approach enables a comprehensive assessment of both global narrative organization and local content preservation.

Each narrative is represented as a directed event graph, where nodes correspond to extracted events (e.g., subject–verb–object triples) and edges encode their temporal progression. Based on this representation, several structural properties are computed, including the number of nodes and edges, graph density, average degree, and longest path.

Structural differences between original and generated graphs are quantified using derived measures such as compression ratios (for nodes and edges), density and degree differences, and changes in longest path length, capturing how the narrative is simplified or reorganized during generation. Additionally, graph edit distance is used as a direct measure of structural dissimilarity, reflecting the minimum number of modifications required to transform one graph into the other.

Complementing these global metrics, event-level alignment is evaluated using precision, recall, and F1-score, based on the overlap between events in the original and generated narratives. These measures assess how accurately key narrative elements are preserved.

All metrics are computed for each story pair and analyzed across the dataset to identify patterns in narrative transformation, including variations across cultural groups and differences between human-authored and machine-generated texts. By jointly considering structural and semantic measures, this framework provides an interpretable and systematic evaluation of narrative fidelity and transformation.

Experimental Results

Overview of the Dataset

The experiments were conducted on a corpus of 30 human-authored folktales collected from three cultural traditions: Persian, German, and Native American, with ten stories in each group. For every original narrative, a corresponding retelling was generated using a large language model, resulting in a parallel dataset of human–machine narrative pairs.

Each narrative was processed through the event extraction pipeline described earlier, producing sequences of subject–verb–object event triples. These events were then organized into directed graphs representing the temporal progression of narrative actions.

Table 1 summarizes the dataset composition and average statistics for the original narratives. Persian stories tend to be longer and contain a higher number of extracted events, leading to more complex graph structures compared to German and Native American narratives. This variation in structural complexity is important when interpreting the results of subsequent comparisons.

Structural Properties of Narrative Graphs

To analyze how narrative structure is modified during retelling, several graph-based metrics were computed, including compression ratios, longest path ratios, and differences in density and average degree.

Table 1 Dataset composition and average number of events, words, and sentences per original narrative in each culture.

Culture	Number of Stories	Avg. Events	Avg. Words	Avg. Sentences
German	10	290.7	2020.0	80.9
Native American	10	225.1	1557.8	73.5
Persian	10	457.6	3280.0	150.1

Table 2 Structural Changes (Average Across Cultures)

Metric	Value
Node Compression Ratio	0.885
Edge Compression Ratio	0.885
Longest Path Ratio	0.885
Density Difference	+0.002
Avg Degree Difference	-0.003

The results indicate a consistent compression effect, where generated narratives contain fewer nodes and edges than the original graphs. On average, LLM retellings preserve approximately 88.5% of the original structural size, suggesting that some events are systematically omitted.

Despite this reduction, the longest path ratio remains stable, indicating that the main sequence of events is largely preserved. Minimal changes in density and average degree further suggest that the local connectivity between events remains consistent.

Table 3 Structural Changes by Culture

Culture	Node Ratio	Edge Ratio	Path Ratio	Density Diff	Degree Diff
German	∼1.004	∼1.00	∼1.00	∼0.002	∼0.000
Native American	∼0.847	∼0.85	∼0.85	∼0.011	∼-0.006
Persian	∼0.803	∼0.80	∼0.80	∼0.004	∼-0.004

Cultural differences are observed in the degree of compression. German narratives show almost no compression, while Persian and Native American stories exhibit stronger reductions in structural complexity. This suggests that LLMs may simplify longer or more complex narratives more aggressively.

Event-Level Preservation

To evaluate how well individual narrative events are preserved, precision, recall, and F1-score were computed.

Table 4 Event Preservation by Culture (Same-Culture Retellings)

Culture	Precision	Recall	F1
German	0.520	0.575	0.542
Native American	0.523	0.479	0.497
Persian	0.543	0.457	0.476

Table 5 Event Preservation Across Conditions

Culture	Cross-Culture F1	Same-Culture F1
German	0.302	0.542
Native American	0.087	0.497
Persian	0.265	0.476

The results show moderate event preservation, with F1-scores ranging from approximately 0.47 to 0.54. Precision values are generally higher than recall, indicating that while generated events are often correct, the model tends to omit a portion of the original events. In addition, Graph edit distance further confirms structural modification, indicating that transformations involve both event removal and minor reordering.

Cross-Cultural Narrative Similarity

Cross-cultural similarity was evaluated using embedding-based cosine similarity between event sequences. Figure 1 visualizes the distribution of similarity scores across all story pairs in the dataset.

Table 6 Average narrative similarity across cultures.

Comparison	Average Similarity
Persian – German	0.5862
Persian – Native American	0.5584
German – Native American	0.5999

The results show moderate similarity values across all cultural pairs (approximately 0.55–0.60), suggesting that narratives share common structural patterns despite cultural differences. This supports the hypothesis that storytelling often follows similar structural principles across traditions, such as sequences of conflict and resolution.

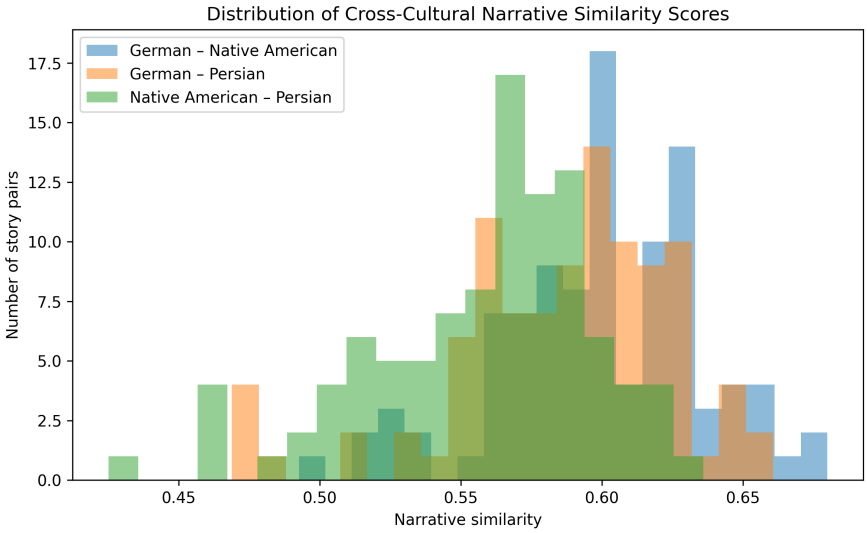

Fig. 1 Distribution of narrative similarity scores.

Comparison Between Human and LLM-Generated Narratives

Combining graph-based and event-level analyses provides a comprehensive view of how LLMs preserve narrative structure.

Overall, LLM-generated narratives remain closer to their original stories than to unrelated narratives from other cultures, as reflected by higher same-culture F1-scores compared to cross-cultural ones. However, structural metrics reveal that this preservation is not complete. Specifically:

1. Compression ratios (0.88) indicate systematic simplification
2. Lower recall values show that some events are omitted
3. Stable path ratios suggest that the main storyline is preserved

These findings indicate that LLMs tend to retain the core narrative backbone while simplifying secondary details.

Conclusion

This study investigated whether narrative structures can be compared across cultures using computational methods, and whether large language models preserve these structures when generating retellings. By representing stories as event-based graphs, the analysis focused on the sequence and organization of narrative actions rather than surface-level text similarity.

The results demonstrate that narratives from different cultural traditions exhibit moderate structural similarity, suggesting the presence of shared storytelling patterns such as sequences of conflict, transformation, and resolution. Despite variations in

length, style, and thematic content, these underlying structures appear to be partially consistent across cultures.

The comparison between human-authored stories and LLM-generated retellings reveals that language models are generally capable of preserving the core narrative backbone. Structural metrics show that the main progression of events is largely maintained, as reflected in stable longest path ratios. However, both graph-based measures and event-level evaluation indicate that this preservation is not complete. Generated narratives consistently exhibit structural compression, with fewer events and lower recall values, suggesting that models tend to omit secondary or less salient narrative elements.

Furthermore, the difference between same-culture and cross-cultural retellings highlights the importance of cultural context. While same-culture generations retain a reasonable level of structural fidelity, cross-cultural retellings show a significant drop in event preservation, indicating that transferring narratives across cultural settings introduces additional distortion.

Overall, the findings suggest that large language models are effective at maintaining the general structure of stories, but tend to simplify narratives by reducing detail. This behavior reflects a balance between coherence and brevity, where the main storyline is preserved while finer narrative elements are compressed or omitted.

Despite these insights, the study has several limitations. The dataset is relatively small and relies on translated narratives, which may obscure certain cultural nuances. Additionally, the event extraction process simplifies complex narrative elements into subject–verb–object structures, potentially overlooking deeper semantic and symbolic aspects of storytelling.

Future work could extend this research by incorporating larger and more diverse corpora, analyzing narratives in their original languages, and applying more advanced event extraction or graph comparison techniques. Exploring different language models and prompting strategies could also provide further insight into how narrative structure is interpreted and reproduced by generative systems.

In conclusion, this work demonstrates that event-based graph representations provide an effective and scalable framework for analyzing narrative structure, enabling meaningful comparison across cultures and between human and machine-generated stories.

AI Usage Statement

AI tools and language models were used in this study for two primary purposes. First, the ChatGPT-4.1 mini model was employed to generate narrative retellings within the same cultural context as well as across different cultural settings. Second, AI assistance was used for language refinement and improving the clarity of the report. All generated content was carefully reviewed, validated, and edited to ensure its accuracy, coherence, and alignment with the objectives of the study.

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